Cancer J. Author manuscript; available in PMC 2024 January 01.

Published in final edited form as:
Cancer J. 2023 ; 29(2): 75–83. doi:10.1097/PPO.0000000000000645.

The microbiome and its impact on allogeneic hematopoietic cell transplantation

Florent Malard[1], **Robert R. Jenq**[2,3]

[1]Sorbonne Université, AP-HP, Centre de Recherche Saint-Antoine INSERM UMRs938, Paris, France ; Service d'Hématologie Clinique et de Thérapie Cellulaire, Hôpital Saint Antoine, AP-HP, Paris, France.

[2]Department of Genomic Medicine, The University of Texas MD Anderson Cancer Center, Houston, TX, USA.

[3]CPRIT Scholar in Cancer Research, Austin, TX 78701, USA.

Abstract

Allogeneic hematopoietic cell transplantation (alloHCT) is a standard curative therapy for a variety of benign and malignant hematological diseases. Previously, patients who underwent alloHCT were at high-risk for complications with potentially life-threatening toxicities, including a variety of opportunistic infections as well as acute and chronic manifestations of graft-versus-host disease (GVHD), where the transplanted immune system can produce inflammatory damage to the patient. With recent advances, including newer conditioning regimens, advances in viral and fungal infection prophylaxis, as well as novel GVHD prophylactic and treatment strategies, improvements in clinical outcomes have steadily improved. One modality with great potential that has yet to be fully realized is targeting the microbiome to further improve clinical outcomes.

In recent years, the intestinal microbiota, which includes bacteria, fungi, viruses, and other microbes that reside within the intestinal tract, has become established as a potent modulator of alloHCT outcomes. The composition of intestinal bacteria, in particular, has been found in large multicenter prospective studies to be strongly associated with GVHD, treatment-related mortality and overall survival. Murine studies have demonstrated a causal relationship between intestinal microbiota injury and aggravated GVHD, and more recently clinical interventional studies of repleting the intestinal microbiota with fecal microbiota transplantation (FMT) have emerged as effective therapies for GVHD.

How the composition of the intestinal bacterial microbiota, which is often highly variable in alloHCT patients, can modulate GVHD and other outcomes is not fully understood. Recent studies, however, have begun to make substantial headway, including identifying particular bacterial subsets and/or bacterial-derived metabolites that can mediate harm or benefit. Here, we review recent studies that have improved our mechanistic understanding of the relationship

---

florent.malard@inserm.fr .

Disclosures:
FM reports honoraria from Therakos/Mallinckrodt, Janssen, Biocodex, Sanofi, JAZZ Pharmaceuticals, Gilead, Novartis, and Astellas. RRJ is on the advisory boards of MaaT Pharma, LisCure, Seres, Kaleido, and Prolacta, has consulted for Merck, Microbiome DX and Karius, and receives patent license fees from Seres.

between the microbiota and alloHCT outcomes, as well as studies that are beginning to establish strategies to modulate the microbiota with the hope of optimizing clinical outcomes.

Introduction to Allogeneic HCT

AlloHCT remains the only curative treatment for patients with a variety of benign and malignant hematological diseases, and the number of alloHCT performed annually has continued to rise[1] despite a transient decrease during the COVID19 pandemic[2]. AlloHCT relies, at least in part, on the immune mediated graft-versus-host tumor or graft-versus-leukemia effect for the eradication of hematologic malignancies. Nevertheless, the counterpart of this effect is the possible recognition and destruction of the recipient's tissue and organs by the donor immune cells, termed GVHD.

GVHD

We distinguish between acute and chronic GVHD (aGVHD and cGVHD) according to disease features. Patients with only signs of aGVHD are classified as classic aGVHD (first episode before day +100 post-transplant), late onset aGVHD (first episode after day +100), recurrent aGVHD, or persistent aGVHD. Patients with any sign of cGVHD are diagnosed as classic cGVHD (only cGVHD features) or overlap cGVHD (presence of both acute and cGVHD features), irrespective of the time of GVHD onset after alloHCT[3].

Acute GVHD usually involves the skin with a maculopapular rash, the lower and/or upper gastrointestinal (GI) tract with diarrhea, nausea and vomiting, and/or the liver with an increased total serum bilirubin level that can lead to jaundice. The pathophysiology of aGVHD can be summed up in a 3-step process[4]. First, the alloHCT conditioning regimen induces tissue damage and release of inflammatory cytokines that lead to activation of patient antigen presenting cells. In the second step, antigen presenting cells activate alloreactive donor T-cells. Finally, in the last step, T-cells migrate to target tissues and cause destruction[4]. Standard first-line treatment for aGVHD is based on steroids (prednisolone 2 mg/kg per day), with a response rate of around 50%[5]. For patients with steroid-refractory aGVHD, ruxolitinib, a Janus kinase (JAK) 2 inhibitor, was recently approved by the Food and Drug Administration (FDA) and the European Medicines Agency (EMA) based on a large phase III randomized clinical trial[6]. Nevertheless, some patients fail on ruxolitinb[7] and treatment of patients with steroid- and ruxolitinib-resistant aGVHD remains an unmet need[4].

Chronic GVHD occurs in 30 to 70% of patients[8] and can involve virtually all organs, in particular, the skin, mouth, eyes, GI tract, liver, lungs, joints, fascia and genital tract[9]. Its pathophysiology is more complex and involves inflammation, T- and B-cell mediated immunity, and fibrosis. First-line treatment relies on steroids alone, prednisolone at 1 mg/kg per day, or combined with a calcineurin inhibitor[10]. Ruxolitinib was also recently approved as second-line treatment in patients with steroid-refractory cGVHD[11].

Infections

AlloHCT is associated with numerous infectious complications related to immunosuppression. We can distinguish three successive phases of immunosuppression[12].

The initial neutropenic phase begins during conditioning and ends with neutrophil recovery. Neutropenia and damage to the GI tract (which constitutes an anatomical barrier) together promote bacterial translocation, which is the primary risk for infection during this phase. In addition, patients also have T- and B-cell immune deficiency as well as functional asplenia. The main pathogens encountered during this phase are Gram-positive and Gram-negative bacteria, *Candida* spp., and herpes simplex virus.

The second phase of immunosuppression begins after neutrophil recovery, up to day +100 after alloHCT. During this phase, infection risk is related to immune deficiency and functional asplenia that can be aggravated by the occurrence of GVHD and its treatment. Cytomegalovirus (CMV), Epstein-Barr virus, adenovirus, BK virus, respiratory syncytial virus, *Pneumocystis jirovecii*, *Candida* spp. and *Aspegillus* spp. are the main causative agents of infection during this phase. Bacterial translocation is also frequent in patients with GI-aGVHD[13].

The late phase starts beyond day +100 after alloHCT and infections occurring during this phase are often linked to the presence of cGVHD. Furthermore, functional asplenia persists, in particular in patients with a history of GVHD. The main infections encountered during this phase are related to encapsulated bacteria (*Streptococcus pneumoniae* and *Haemophilus influenzae*), *Aspergillus* spp., Varicella-zoster virus or Epstein-Barr virus that can lead to a post-transplantation lymphoproliferative disorder.

Relapse

The leading cause of alloHCT failure remains the risk of relapse of the underlying malignancy, and this is particularly relevant in acute myeloid leukemia (AML), the most common indication for alloHCT. Despite the latter providing the highest potential for long-term survival in such patients[14], ultimately up to half of AML patients will relapse, depending on the disease characteristics and the type of conditioning regimen. Furthermore, outcomes for those who relapse after alloHCT is very poor, with a 2-year overall survival (OS) of less than 20%[15]. Therefore, post-transplant maintenance is being developed for patients with negative minimal residual disease after alloHCT. Maintenance treatment modalities include prophylactic donor lymphocyte infusion[16] as well as pharmacological intervention with hypomethylating agents[17] or targeted therapy including Fms-like tyrosine kinase 3 (FLT3) or isocitrate dehydrogenase inhibitors[18].

The microbiome and alloHCT

Early studies of the microbiome in alloHCT

Early experiments in mouse models performed in the 1970s demonstrated that the microbiome, including in particular intestinal bacteria, contributed to aGVHD pathophysiology. Germ-free mice maintained in gnotobiotic isolators were found to have very little aGVHD after undergoing alloHCT[19], and similar findings were later seen when normal mice were given a cocktail of oral antibiotics that substantially reduced microbial colonization of the intestinal tract[20]. Subsequently, improved outcomes of patients transplanted in protective isolation conditions, which included a cocktail of oral antibiotics as well as other measures[21], were reported, including reduced aGVHD[22], though this finding was notably observed prior to the development of use of calcineurin inhibitors for aGVHD prophylaxis.

Microbiome deep sequencing in alloHCT

With the advent of deep sequencing methods such as 16S ribosomal RNA gene sequencing in the late 2000s, it became possible to quickly and affordably characterize the bacterial composition of patient biospecimens collected as they undergo the alloHCT process. One of the first initial observations was that alloHCT patients often experience substantial changes in the composition of their intestinal bacteria[23], and these perturbations are associated with poor outcomes, including increased rates of bloodstream infections[24], aGVHD[25-27] and mortality[24, 26, 27]. These findings have since been confirmed in large multi-center studies[28].

Interestingly, not all manifestations of aGVHD are associated with changes in intestinal bacteria; while development of lower GI-aGVHD is strongly associated, aGVHD that involves only the skin[27] or upper gut[29] are not strongly associated with microbiome changes in the lower intestinal tract. This observation suggests that perhaps the local microbiome adjacent to the afflicted tissue may be particularly impactful on aGVHD biology. Studies of the skin microbiome[30, 31] found that cutaneous aGVHD was associated with increased abundances of *Staphylococcus*, a finding highly reminiscent of that seen in children with atopic dermatitis[32]. Pulmonary toxicities and changes in the pulmonary microbiome have also been found to be co-associated in pediatric alloHCT patients[33, 34]. Changes in the ocular microbiota have also been observed in patients with ocular cGVHD, including in composition of bacteria as well as viruses[35]. Whether the microbiome of the oral cavity or upper gut could be similarly associated with upper GI-aGVHD has not yet been well studied, though the dental biofilm microbiome has been reported to be associated with aGVHD[36].

Sources of microbiome disruption in alloHCT

Why the microbiome is so frequently perturbed in alloHCT patients can been explained to a large degree by the common and necessary use of antibiotics, which in turn can lead to worsened clinical outcomes (Figure 1). These are given as bacterial prophylaxis, as empiric treatment for neutropenic fever, and as treatment for documented infections. Interestingly, while several antibiotics for each of these clinical settings are available and considered by experts to be largely clinically equivalent, the impact on the intestinal microbiome is not identical. In the prophylactic setting, metronidazole, when given in addition to ciprofloxacin, has been reported to be detrimental to the microbiome as well as associated with worsened clinical outcomes including GI-aGVHD and treatment-related mortality[37, 38]. Rifaximin prophylaxis was associated with improved outcomes but, to our knowledge, has not yet been prospectively compared to prophylaxis with fluoroquinolones alone.

In the setting of empiric treatment of neutropenic fever and documented infections, choice of antibiotic has been found to be associated with microbiome disruption and toxicity outcomes[39, 40]. Antibiotics that have high activity against obligate anaerobes such as carbapenems have been particularly strongly associated with harm. Murine models of aGVHD have recapitulated these observations, with carbapenems seen to markedly aggravate aGVHD, especially lower GI-aGVHD. Results of prospective clinical trials evaluating choice of antibiotic and clinical outcomes have not yet been reported but developing strategies that adequately prevent and treat infections while sparing the microbiome as much as feasibly possible is an appealing approach.

In addition to antibiotics, HCT conditioning is another common perturbator of the intestinal microbiome, with a direct correlation between HCT conditioning intensity and microbiome injury in clinical studies[41]. An examination of the intestinal microbiome of patients undergoing alloHCT conditioning prior to development of neutropenic fever, as well as the microbiome of mice following either total body irradiation (TBI) or treatment with melphalan in the absence of antibiotic treatment, both demonstrated that HCT conditioning alone can markedly modulate the composition of intestinal bacteria[42]. Interestingly, murine experiments demonstrated that mice, similar to humans, consume less food after TBI, and that much of the effects of HCT on the microbiome can be recapitulated with a simple 50% reduction in oral food intake in unirradiated mice[42]. Hence nutritional supplementation strategies represent an approach that could mediate meaningful benefits on the intestinal microbiome and in turn improve clinical outcomes. Some studies have already begun to test this hypothesis, including administration of galactooligosaccharides which were shown to produce a substantial benefit in mice with aGVHD[43] and a Phase I dose-escalation study of fructooligosaccharides[44] which indicated a potential benefit by producing an increasing trend in regulatory T cells in patients. Please see below for additional studies of nutritional interventions.

Avoiding certain nutrients may be another approach to sculpt the microbiome in a way to improve clinical outcomes. Dietary lactose, which is commonly found in dairy products, has been found to be an important contributor to expansion of *Enterococcus*, which can in turn aggravate aGVHD in both patients and in mouse models[45]. Interestingly, host genetic polymorphisms were also found to play a role, with patients carrying lactose-nonabsorber genotypes shown to have a propensity to increased abundances of *Enterococcus*, demonstrating an interplay between the diet, host genetics, the microbiome and clinical outcomes.

GVHD-modulating bacterial subsets

Lactobacillus strains have been found to experimentally mediate a benefit in murine models of aGVHD[25, 46], potentially by inhibiting the expansion of enterococci. A small clinical trial of administering lactobacilli to patients, however, did not replicate this finding[47]. Another potentially beneficial subset of bacteria includes Clostridia, which encompass *Eubacterium*, *Faecalibacterium*, *Roseburia*, *Ruminococcus* and *Blautia* species, have been clinically associated with protection from aGVHD in patients[27, 39]. Additionally, depleting Clostridia in mice aggravates aGVHD[39, 40] while administering Clostridia can reduce aGVHD severity[48], providing a rationale for strategies to introduce Clostridia to patients at risk for aGVHD.

Subsets of bacteria linked to aggravated aGVHD have also been identified, though are better studied in mouse models than in patients. These include *Enterococcus*[45], as described above, and *E. coli*[49]. Interestingly, suppressing *E. coli* with oral polymyxin B reduced aGVHD substantially in mice, a finding reminiscent of the early studies of mice treated with nonabsorbable antibiotics[20]. However, a recent small randomized clinical trial of 20 patients evaluated the effects of oral vancomycin and polymyxin B and found that this antibiotic regimen produced a trend towards reduced bloodstream infections (5 vs 1 patients, p=0.09), but no difference in aGVHD incidence (p=0.6)[50]. This could indicate that such a broadly decontaminating strategy may not be ideal for aGVHD prevention, and instead narrowly targeting *E. coli* and other Gammaproteobacteria could be worthy of examining clinically.

We recently found that *Bacteroides thetaiotaomicron* (Bt), which can be found in the intestine of mice as well as humans and achieves high abundances after meropenem therapy, can aggravate aGVHD in mice after treatment with meropenem[40]. Bt is known to be capable of degrading colonic mucus, which is composed of glycosylated mucin proteins that form a protective layer that functions to physically separate bacteria from the intestinal epithelium. Meropenem treatment led to substantial thinning of colonic mucus, and also led to upregulation of mucus-degrading enzymes by Bt. Interestingly, an investigation of monosaccharide concentrations in the colonic lumen showed that several monosaccharides became depleted after meropenem treatment, perhaps reflecting a loss of commensal bacteria that normally function to degrade dietary fibers and starches into monosaccharides. We wondered if these changes in monosaccharide concentrations could be modifying Bt behavior, and found that cultivating Bt in the presence of xylose led to significant downregulation of mucus-degrading enzymes, as well as reduced consumption of mucin by Bt. We the evaluated administration of oral xylose to mice and found that this could prevent mucus-degradation and improve survival in mice with aGVHD, thus preventing the harmful effects of meropenem treatment. These results demonstrated that an adaptive response of certain intestinal bacteria to an altered environment in the intestine can aggravate GI-aGVHD, and a better understanding of this behavior can lead to the development of targeted strategies to restore an environment that is less pro-inflammatory.

Bacterial metabolites that modulate aGVHD

Additional mechanistic studies have identified in mouse models specific bacterial metabolites that can modulate aGVHD severity. Potentially protective compounds include butyrate, which is produced by bacterial metabolism of carbohydrates[48] and indoles, which are produced by bacterial metabolism of tryptophan and can signal through the aryl hydrocarbon receptor (AHR)[51]. Potentially harmful compounds include trimethylamine N-oxide, a product of bacterial choline metabolism[52]. Supporting the translational significance of these experimental findings are studies of serum metabolites in the setting of clinical alloHCT that have found that concentrations of indoles, as well as bile acids which are strongly modulated by intestinal bacterial metabolism, are disrupted in patients with clinical aGVHD[53].

Microbiome and outcomes beyond GI-aGVHD

One question that arises is whether the association in alloHCT patients between survival and an intact intestinal microbiome is driven primarily by GI-aGVHD, or if other aspects of transplantation biology are also being modulated. The answer is not yet fully clear, but a body of literature suggests that immune recovery may be strongly modulated by the intestinal microbiome. A recent study of the microbiome in alloHCT patients who received antithymocyte globulin-based conditioning, and thus had a low rate of severe aGVHD, found that while diversity of the intestinal flora at the engraftment period was not associated with aGVHD incidence, it was nevertheless an independent predictor of longer survival[54], with more infection-related deaths seen in low-diversity patients.

Bloodstream infections were one of the first infection-related outcomes to be associated with microbiome composition, though the underlying process was thought to be related to intestinal expansion and translocation rather than modulation of immune recovery[24]. An association between increased butyrate-producing bacteria and reduced viral pneumonias suggested that immune cell recovery could also be microbiome-modulated[55]. Studies examining for associations between intestinal microbiome composition and leukocyte recovery[56], as well as T cell recovery[57], have found strong associations between microbiome composition and better immune recovery. Further supporting evidence include a mouse study where depletion of the intestinal microbiota resulted in impaired recovery of lymphocyte and neutrophil counts, while recovery of the hematopoietic stem and progenitor compartments and the erythroid lineage were largely unaffected[58]. Interestingly, this could be addressed in mice with a simple nutritional intervention of sucrose supplementation. In patients, autologous FMT has been found to be effective in restoring the microbiome composition in patients after alloHCT[59] and can accelerate leukocyte count recovery[56].

Finally, a study of the intestinal microbiome in patients after autologous HCT, a procedure where there is essentially no risk for GVHD, found that microbiome diversity at neutrophil engraftment was associated with improved overall survival[60]. Given that this patient population is typically at highest risk for cancer progression with very little risk for treatment-related mortality, these results suggested that the intestinal microbiome composition could be modulating cancer response to HCT conditioning. Similar microbiome associations with cancer response have also been seen in the alloHCT[61], though possible underlying mechanisms remain unclear.

Microbiome associations with neutropenic fever

Because a common cause of a perturbed microbiome in alloHCT patients is often the use of antibiotics as empiric treatment of neutropenic fever, an examination of causes of neutropenic fever is warranted. It is not well-understood why some patients will develop neutropenic fever and others do not. Interestingly, severity and duration of neutropenia are not strongly associated with developing fever; rather the severity of mucositis is more predictive[62]. How mucositis leads to fever, however, is unclear. Of those who develop fever, 10-20% will be diagnosed with bacteremia, most commonly with gastrointestinal commensal bacteria such as Gammaproteobacteria or Bacilli (including enterococci and streptococci), while the large majority will not have an infection source identified[63].

Given that the majority of bloodstream infections identified from patients with febrile neutropenia arise from an intestinal origin, we recently asked if the composition of intestinal bacteria could be associated with development of fever in HCT patients when they become neutropenic[42]. We found that patients who later developed fever had increased relative abundances of *Bacteroides* and *Akkermansia muciniphila* in their fecal samples collected at onset of neutropenia. Interestingly, in febrile patients these two types of bacteria also tended to expand when compared to baseline, especially *Akkermansia muciniphila*, suggesting that chemotherapy conditioning led to an increase in these bacteria. Evaluating the microbiome of mice following TBI or melphalan similarly showed expansions in abundances of *Bacteroides* and *Akkermansia muciniphila*. Both of these types of bacteria are commonly capable of degrading mucins, and we found that TBI or melphalan resulted in substantial thinning of the colonic mucus layer. *Akkermansia muciniphila* was the predominant mucus degrader in mice, and was found to be regulated by propionate, a short chain fatty acid bacterial metabolite that decreases in concentration in the colon in individuals that are not eating normally. Oral administration of propionate to mice after TBI led to reduced hypothermia, improved intestinal barrier function, and reduced colonic inflammation, suggesting that restoring levels of certain bacterial metabolites can also help improve intestinal homeostasis and may be a strategy to prevent clinical neutropenic fever (Figure 2).

Microbiome-based interventional strategies

Given the impact of the microbiome on alloHCT outcomes, it seems important to evaluate the available strategies to modulate the microbiota and improve patient outcomes. Available approaches include diet modification and prebiotic administration, probiotics and in particular, fecal microbiota transplantation, and refined antibiotic treatment.

Prebiotics are non-digestible food ingredients that stimulate growth and metabolism of beneficial bacteria in the gut[64]. Thus, dietary changes including in particular fiber supplementation, or administration of prebiotics, can beneficially change the composition of the intestinal microbiota[65-67]. To be effective, however, identifying key beneficial symbiotic bacterial strains will be indispensable in order to rebuild a healthy microbiota. In the setting of alloHCT and particularly in patients with aGVHD, given the high prevalence, high degree, and harmful impact of dysbiosis, dietary modification and prebiotic supplementation alone are unlikely to be sufficient, and concomitant administration of probiotics will likely be needed to restore a healthy microbiota in these patients.

Probiotics are live micro-organisms that can be administered alone or as a consortium of selected symbiotic strains. The gut microbiota typically consists of hundreds of distinct microbial species, and so designing a consortium of symbiotic microbial species with sufficient breadth that can functionally recapitulate a full gut microbiota, in addition to well-colonize and displace the existing gut microbiota, can be challenging[68]. Fecal microbiota transplantation (FMT), consisting of the transfer of an entire fecal microbiota, is thus being developed for numerous clinical indications[23].

FMT consists of the transfer of a complex community of microorganisms from a healthy donor to a recipient. Thorough screening of the donor is critically important and strictly regulated, based on recommendations from scientific societies and mandatory requirements from health authority agencies, in order to sufficiently minimize the risk of introducing potential pathogens in the FMT product[69]. These mandatory guidelines were reinforced after a report of antibiotic-resistant *Escherichia coli* bacteremia transmitted by FMT in two immunosuppressed patients (one patient with cirrhosis of the liver, one patient with a history of alloHCT), leading to the death of one patient[70]. FMT aims to restore the beneficial function of a healthy microbiota and the crosstalk with the immune system. FMT is a proven treatment for prevention of a second or further recurrence of *Clostridioides difficile* (*C. difficile*) infection (CDI)[71], based on a randomized clinical trial that demonstrated its superiority compared to conventional antibiotics[72]. Furthermore, it was recently shown that FMT is superior to standard of care vancomycin in patients with first and second CDI in achieving a sustained resolution[73]. Use of FMT for CDI was found to be safe and effective in a cohort of 80 immunocompromised patients[74]. Numerous clinical trials have investigated FMT for a large spectrum of clinical indications, including ulcerative colitis and Crohn's disease, multidrug resistant infection, liver diseases, obesity and other metabolic syndromes, neuropsychiatric disorders, and oncologic settings, including hematological malignancies[68]. In addition to prebiotics, probiotics and FMT, refinement of antibiotics would seem most important in the setting of alloHCT in order to decrease their impact on gut microbiota diversity, given the impact of antibiotics on patients' outcomes after alloHCT[38, 75-77].

Current cutting-edge clinical investigations

Nutrition support and prebiotics

Impaired nutritional status has been identified as a risk factor for severe aGVHD[78] and since oral intake is severely reduced early after alloHCT, nutritional support is indispensable. A recent meta-analysis showed that enteral nutrition (EN) was associated with a lower rate of aGVHD, including severe grade III-IV aGVHD and GI-aGVHD, compared to parenteral nutrition (PN)[79]. Furthermore, a study evaluated the impact of the route of nutrition on gut microbiota composition after alloHCT[80] in 23 adult patients who were randomly assigned to receive EN or PN. While there was no difference in gut microbial diversity at day +30, there were significant differences in the microbial composition with a higher abundance of taxa associated with increased short chain fatty acid (SCFA) production in patients that received EN. On the contrary, the PN group showed a greater abundance of *Enterococcus* and Gammaproteobacteria, such as *Klebsiella*. Similarly, 20 pediatric alloHCT patients receiving EN had less gut microbiota injury and an almost complete recovery of microbial diversity after alloHCT[81]. In contrast, patients receiving PN had more gut microbiota dysbiosis and never achieved a return to pre-transplant microbial status. Overall, when oral intake is insufficient, the use of EN seems to preserve gut microbiota composition and is associated with improved clinical outcome. Therefore, EN should be used as the first line nutritional support, as is recommended by international guidelines[82].

Several studies have also investigated prebiotics in the setting of alloHCT. Iyama *et al.* evaluated the impact of glutamine, fiber and oligosaccharides (GFO) on GI mucosal injury among 22 alloHCT recipients[83]. GFO was delivered orally 3 times per day starting 7 days prior to the start of conditioning and continued until day +28 after alloHCT. These patients were compared with a matched-pair control group of 22 alloHCT patients that did not receive GFO supplementation. Administration of GFO was associated with a decrease severity of the intestinal mucositis and a higher OS, while there was no effect on GVHD.

GFO was then evaluated in combination with resistant starch (RS) in a prospective study that included 49 alloHCT patients[84]. RS is a prebiotic that increases the amount of SCFAs[85], including butyrate and the abundance of butyrate-producing bacteria in the intestine[86]. RS and GFO were administered from the start of the conditioning regimen until day +28 after alloHCT. When compared with a historical control group of 142 patients that did not received RS and GFO, patients that received the prebiotic had less mucosal injury and a reduced incidence of all grade and grade II-IV aGVHD. Furthermore, they found that the gut microbial diversity was well maintained, and the butyrate-producing bacterial population was better preserved by prebiotic intake. Another prebiotic, fructo-oligosaccharides (FOS) is thought to support the growth of commensal organisms such as *Bifidobacterium* and to increase levels of SCFAs. FOS was evaluated through a phase 1 dose-escalation clinical trial that compared 15 patients undergoing reduced-intensity alloHCT with 16 concurrent controls[44]. FOS at 10g per day was safe and well-tolerated and furthermore, FOS patients had a significantly different community-level gut microbiota composition on the day of alloHCT compared to controls, though this effect was not sustained after alloHCT.

Fecal microbiota transplantation

Regarding FMT in the setting of alloHCT, clinical investigation has mainly focused on the treatment of steroid-dependent (SD) or steroid-resistant (SR) GI-aGVHD. Among several case reports and small series, van Lier *et al*. reported outcomes of 15 patients with SR- (n=6) or SD- (n=9) GI-aGVHD that received FMT with fresh feces from a single donor via a nasoduodenal tube[87]. Complete response (CR) was observed at 1 month after FMT in 10 of 15 patients, including 3 of 6 patients with SR-GI-aGVHD and 7 of 9 patients with SD-aGVHD. Importantly, patients that achieved a CR at 1 month exhibited a higher gut microbial α-diversity, partial but demonstrable engraftment of donor bacterial species, and an increased abundance of butyrate-producing bacteria, including Clostridiales and *Blautia* species.

A prospective phase IIa clinical trial evaluated FMT in 24 patients with SR-GI-aGVHD[88]. The FMT product was manufactured under current good manufacturing process (cGMP) guidelines using feces from a pool of healthy donors. It was characterized by a highly consistent richness of 455 ±3% operational taxonomic units (OTUs). At day +28, the GI-overall response rate (ORR) was 38%, including 5 CR, 2 very good partial response (VGPR) and 2 partial response (PR). Similar to the previous study, fecal bacterial microbiota α-diversity was significantly higher in patients who responded at day +28 compared to non-responding patients. Responders also demonstrated a higher engraftment proportion with the FMT product compared to non-responding patients. In a follow-up study, 52 patients were treated with the same FMT product as part of a compassionate use/expanded access program (EAP)[88]. In this cohort, all patients had SR- (n=43) or SD- (n=9) GI-aGVHD and had received a median of 3 (range, 1-6) lines of treatment for aGVHD, including ruxolitinib for 40 patients. The GI-ORR at day +28 was 58% (17 CR, 9 VGPR, and 4 PR).

A meta-analysis recently evaluated the efficacy and safety of FMT for the treatment of SR- or SD-GI-aGVHD[89]. This meta-analysis included 23 studies: 2 prospective cohort studies, 10 prospective single arm studies, 2 retrospective single arm studies, 2 case series and 7 case reports. Among 242 patients, 100 achieved a CR, 61 a PR, and 81 had no response after FMT. The pooled clinical remission rate was 64% (51-77%) in prospective single arm studies and 81% (62-95%) in retrospective studies, case series and case reports. Only 5 (2.1%) patients had FMT related infection events and all recovered after treatment.

Overall, these studies and meta-analysis confirmed that FMT is a safe and promising treatment for GI-aGVHD, nevertheless prospective control studies are still needed before it becomes standard of care. Furthermore, it would be important to evaluate in such studies the efficacy of FMT on others organ involved in aGVHD, namely the skin and liver.

Two studies have also shown that FMT after alloHCT and neutrophil recovery is safe and restores gut microbial diversity. One was a phase II randomized control trial that compared autologuous FMT (n=14) *versus* no intervention (n=11)[59]. Autologuous FMT grafts were collected immediately before alloHCT hospitalization. Compared to the control group, autologuous FMT increased microbial diversity and reestablished the intestinal microbiota composition that existed before alloHCT and use of large spectrum antibiotics. In the second study, 13 patients received 15 FMT capsules per day for 2 consecutive days up to a total of 30 capsules, at a median of 27 days (range, 19-45) after alloHCT[90]. After FMT, they also found an improvement in intestinal microbiome diversity and an expansion of donor-stool taxa. Two of the 13 patients developed subsequent GI-aGVHD.

Antibiotics

In an early study performed in the late 1990s, use of ciprofloxacin plus metronidazole was associated with a reduced incidence of grade II-IV aGVHD compared to patients that received ciprofloxacin alone[91] as a strategy to suppress anaerobic bacterial overgrowth[92]. More recent studies, however, indicate that metronidazole appears to increase the risk for aGVHD[38, 93]. One potential explanation for this discrepancy could be due to changes in antibiotic resistance patterns over time. Indeed, one study of gut decontamination found that while this strategy can be effective, the risk of aGVHD significantly increases in patients where antibiotics are ineffective in suppressing growth of intestinal microbes[94].

Regarding use of curative antibiotics, one study found a detrimental impact on alloHCT outcomes after early use of broad spectrum antibiotics[77] and more particularly in a study of broad-spectrum antibiotics with anaerobic activity such as piperacillin-tazobactam or imipenem-cilastatin[75]. Therefore, while it is impossible to avoid use of such antibiotics for febrile neutropenia during alloHCT, we recommend following the guidelines published by the Expert Group of the 4th European Conference on Infections in Leukemia (ECIL) that can help decrease antibiotic burden, and therefore their detrimental effect on gut microbiota diversity[95]. In particular, first-line treatment with carbapenems should be avoided in patients with no particular risk factors. In addition, for neutropenic patients with fever of unknown origin who are hemodynamically stable since presentation and afebrile for at least 48 hours, discontinuation of antibiotics after 72 hours or later can be considered even prior to neutrophil recovery, given the evidence for the safety of this approach[96], including with carbapenem treatment[97].

Conclusion

Over the last decade, the association between the gut microbiota and patients' outcomes after alloHCT, including infectious complications, acute and cGVHD, risk of relapse of the underlying malignancy, non-relapse mortality and OS, have been clearly established. Importantly, in addition to the bacterial gut microbiota, other microbiomes were also found to have an impact on alloHCT outcome, not only fungal and viral gut microbiota, but also more recently, the skin microbiota. Mechanisms underlying this association have begun to be identified, in particular through bacterial metabolites, effects of nutrition, and effects on intestinal mucus. Based on these findings, development of strategies to manipulate the gut microbiota to prevent dysbiosis appears to be very important in improving patients' outcomes. Up to now, strategies have focused on the transplantation of a high diversity bacterial microbiome from a single donor or a pool of donors, but development of a refined gut microbiota manipulation with transplantation of a pool of relevant selected strains deserves investigation. In addition, administration of probiotics or selected nutrition is currently being investigated and will probably be an indispensable adjuvant therapy to complete the effectiveness of probiotics. Finally, while gut microbiota manipulation in alloHCT is in its infancy, we nevertheless expect that such prebiotic/probiotic-based strategies will one day became standard of care for alloHCT patients, similar to those of immunosuppressive drugs or anti-infection prophylaxis.

Sources of funding:

We would like to acknowledge funding from the National Institutes of Health (R01HL124112 and P30CA016672 to RRJ).

References

1. Passweg JR, Baldomero H, Chabannon C, et al. Hematopoietic cell transplantation and cellular therapy survey of the EBMT: monitoring of activities and trends over 30 years. Bone Marrow Transplant. 2021.
2. Passweg JR, Baldomero H, Chabannon C, et al. Impact of the SARS-CoV-2 pandemic on hematopoietic cell transplantation and cellular therapies in Europe 2020: a report from the EBMT activity survey. Bone Marrow Transplant. 2022;57: 742–752. [PubMed: 35194156]
3. Filipovich AH, Weisdorf D, Pavletic S, et al. National Institutes of Health consensus development project on criteria for clinical trials in chronic graft-versus-host disease: I. Diagnosis and staging working group report. Biol Blood Marrow Transplant. 2005;11: 945–956. [PubMed: 16338616]
4. Malard F, Huang XJ, Sim JPY. Treatment and unmet needs in steroid-refractory acute graft-versus-host disease. Leukemia. 2020;34: 1229–1240. [PubMed: 32242050]
5. MacMillan ML, DeFor TE, Weisdorf DJ. The best endpoint for acute GVHD treatment trials. Blood. 2010;115: 5412–5417. [PubMed: 20388871]
6. Zeiser R, von Bubnoff N, Butler J, et al. Ruxolitinib for Glucocorticoid-Refractory Acute Graft-versus-Host Disease. N Engl J Med. 2020;382: 1800–1810. [PubMed: 32320566]
7. Mohty M, Holler E, Jagasia M, et al. Refractory acute graft-versus-host disease: a new working definition beyond corticosteroid refractoriness. Blood. 2020;136: 1903–1906. [PubMed: 32756949]
8. Lee SJ, Flowers MED. Recognizing and Managing Chronic Graft-Versus-Host Disease. Hematology. 2008;2008: 134–141.
9. Jagasia MH, Greinix HT, Arora M, et al. National Institutes of Health Consensus Development Project on Criteria for Clinical Trials in Chronic Graft-versus-Host Disease: I. The 2014 Diagnosis

and Staging Working Group report. Biol Blood Marrow Transplant. 2015;21: 389–401.e381. [PubMed: 25529383]

10. Penack O, Marchetti M, Ruutu T, et al. Prophylaxis and management of graft versus host disease after stem-cell transplantation for haematological malignancies: updated consensus recommendations of the European Society for Blood and Marrow Transplantation. Lancet Haematol. 2020;7: e157–e167. [PubMed: 32004485]
11. Zeiser R, Polverelli N, Ram R, et al. Ruxolitinib for Glucocorticoid-Refractory Chronic Graft-versus-Host Disease. N Engl J Med. 2021;385: 228–238. [PubMed: 34260836]
12. Apperley J, Carreras E, Gluckman E, Masszi T. The EBMT Handbook, Haematopoietic Stem Cell Transplantation. 2012 Revised Edition ed: EBMT, 2012.
13. Mori Y, Yoshimoto G, Nishida R, et al. Gastrointestinal Graft-versus-Host Disease Is a Risk Factor for Postengraftment Bloodstream Infection in Allogeneic Hematopoietic Stem Cell Transplant Recipients. Biol Blood Marrow Transplant. 2018;24: 2302–2309. [PubMed: 29909153]
14. Dohner H, Estey E, Grimwade D, et al. Diagnosis and management of AML in adults: 2017 ELN recommendations from an international expert panel. Blood. 2017;129: 424–447. [PubMed: 27895058]
15. Bazarbachi A, Schmid C, Labopin M, et al. Evaluation of Trends and Prognosis Over Time in Patients with AML Relapsing After Allogeneic Hematopoietic Cell Transplant Reveals Improved Survival for Young Patients in Recent Years. Clinical Cancer Research. 2020;26: 6475–6482. [PubMed: 32988970]
16. Schmid C, Schleuning M, Ledderose G, Tischer J, Kolb H-J. Sequential Regimen of Chemotherapy, Reduced-Intensity Conditioning for Allogeneic Stem-Cell Transplantation, and Prophylactic Donor Lymphocyte Transfusion in High-Risk Acute Myeloid Leukemia and Myelodysplastic Syndrome. Journal of Clinical Oncology. 2005;23: 5675–5687. [PubMed: 16110027]
17. Guillaume T, Malard F, Magro L, et al. Prospective phase II study of prophylactic low-dose azacitidine and donor lymphocyte infusions following allogeneic hematopoietic stem cell transplantation for high-risk acute myeloid leukemia and myelodysplastic syndrome. Bone Marrow Transplant. 2019;54: 1815–1826. [PubMed: 31089280]
18. Battipaglia G, Ruggeri A, Massoud R, et al. Efficacy and feasibility of sorafenib as a maintenance agent after allogeneic hematopoietic stem cell transplantation for Fms-like tyrosine kinase 3-mutated acute myeloid leukemia. Cancer. 2017;123: 2867–2874. [PubMed: 28387928]
19. Jones JM, Wilson R, Bealmear PM. Mortality and gross pathology of secondary disease in germfree mouse radiation chimeras. Radiat Res. 1971;45: 577–588. [PubMed: 4396814]
20. van Bekkum DW, Roodenburg J, Heidt PJ, van der Waaij D. Mitigation of secondary disease of allogeneic mouse radiation chimeras by modification of the intestinal microflora. J Natl Cancer Inst. 1974;52: 401–404. [PubMed: 4150164]
21. Bodey GP. The changing face of febrile neutropenia-from monotherapy to moulds to mucositis. Fever and neutropenia: the early years. J Antimicrob Chemother. 2009;63 Suppl 1: i3–13. [PubMed: 19372179]
22. Storb R, Prentice RL, Buckner CD, et al. Graft-versus-Host Disease and Survival in Patients with Aplastic Anemia Treated by Marrow Grafts from HLA-Identical Siblings. New England Journal of Medicine. 1983;308: 302–307. [PubMed: 6337323]
23. Ubeda C, Taur Y, Jenq RR, et al. Vancomycin-resistant Enterococcus domination of intestinal microbiota is enabled by antibiotic treatment in mice and precedes bloodstream invasion in humans. J Clin Invest. 2010;120: 4332–4341. [PubMed: 21099116]
24. Taur Y, Xavier JB, Lipuma L, et al. Intestinal domination and the risk of bacteremia in patients undergoing allogeneic hematopoietic stem cell transplantation. Clin Infect Dis. 2012;55: 905–914. [PubMed: 22718773]
25. Jenq RR, Ubeda C, Taur Y, et al. Regulation of intestinal inflammation by microbiota following allogeneic bone marrow transplantation. J Exp Med. 2012;209: 903–911. [PubMed: 22547653]
26. Holler E, Butzhammer P, Schmid K, et al. Metagenomic Analysis of the Stool Microbiome in Patients Receiving Allogeneic Stem Cell Transplantation: Loss of Diversity Is Associated with Use

of Systemic Antibiotics and More Pronounced in Gastrointestinal Graft-versus-Host Disease. Biol Blood Marrow Transplant. 2014;20: 640–645. [PubMed: 24492144]

27. Jenq RR, Taur Y, Devlin SM, et al. Intestinal Blautia is associated with reduced death from graft-versus-host disease. Biology of Blood and Marrow Transplantation. 2015;21: 1373–1383. [PubMed: 25977230]
28. Peled JU, Gomes ALC, Devlin SM, et al. Microbiota as Predictor of Mortality in Allogeneic Hematopoietic-Cell Transplantation. N Engl J Med. 2020;382: 822–834. [PubMed: 32101664]
29. Burgos da Silva M, Ponce DM, Dai A, et al. Preservation of fecal microbiome is associated with reduced severity of Graft-versus-Host Disease. Blood. 2022.
30. Gu Y, Sun J, Li K, Wu X, Zhang J. Alteration in the Skin Microbiome in Cutaneous Graft Versus Host Disease. Acta Derm Venereol. 2021;101: adv00374. [PubMed: 32812057]
31. Bayer N, Hausman B, Pandey RV, et al. Disturbances in microbial skin recolonization and cutaneous immune response following allogeneic stem cell transfer. Leukemia. 2022.
32. Kong HH, Oh J, Deming C, et al. Temporal shifts in the skin microbiome associated with disease flares and treatment in children with atopic dermatitis. Genome Res. 2012;22: 850–859. [PubMed: 22310478]
33. Zinter MS, Lindemans CA, Versluys BA, et al. The pulmonary metatranscriptome prior to pediatric HCT identifies post-HCT lung injury. Blood. 2021;137: 1679–1689. [PubMed: 33512420]
34. Zinter MS, Versluys AB, Lindemans CA, et al. Pulmonary microbiome and gene expression signatures differentiate lung function in pediatric hematopoietic cell transplant candidates. Sci Transl Med. 2022;14: eabm8646. [PubMed: 35263147]
35. Li J, Liang Q, Huang F, et al. Metagenomic Profiling of the Ocular Surface Microbiome in Patients After Allogeneic Hematopoietic Stem Cell Transplantation. Am J Ophthalmol. 2022;242: 144–155. [PubMed: 35551905]
36. Heidrich V, Bruno JS, Knebel FH, et al. Dental Biofilm Microbiota Dysbiosis Is Associated With the Risk of Acute Graft-Versus-Host Disease After Allogeneic Hematopoietic Stem Cell Transplantation. Front Immunol. 2021;12: 692225. [PubMed: 34220852]
37. Weber D, Oefner PJ, Dettmer K, et al. Rifaximin preserves intestinal microbiota balance in patients undergoing allogeneic stem cell transplantation. Bone Marrow Transplant. 2016;51: 1087–1092. [PubMed: 26999466]
38. Weber D, Hiergeist A, Weber M, et al. Detrimental Effect of Broad-spectrum Antibiotics on Intestinal Microbiome Diversity in Patients After Allogeneic Stem Cell Transplantation: Lack of Commensal Sparing Antibiotics. Clin Infect Dis. 2019;68: 1303–1310. [PubMed: 30124813]
39. Shono Y, Docampo MD, Peled JU, et al. Increased GVHD-related mortality with broad-spectrum antibiotic use after allogeneic hematopoietic stem cell transplantation in human patients and mice. Sci Transl Med. 2016;8: 339ra371.
40. Hayase E, Hayase T, Jamal MA, et al. Mucus-degrading Bacteroides link carbapenems to aggravated graft-versus-host disease. Cell. 2022;185: 3705–3719 e3714. [PubMed: 36179667]
41. Shouval R, Waters NR, Gomes ALC, et al. Conditioning regimens are associated with distinct patterns of microbiota injury in allogeneic hematopoietic cell transplantation. Clin Cancer Res. 2022.
42. Schwabkey ZI, Wiesnoski DH, Chang CC, et al. Diet-derived metabolites and mucus link the gut microbiome to fever after cytotoxic cancer treatment. Sci Transl Med. 2022;14: eabo3445. [PubMed: 36383683]
43. Holmes ZC, Tang H, Liu C, et al. Prebiotic galactooligosaccharides interact with mouse gut microbiota to attenuate acute graft-versus-host disease. Blood. 2022;140: 2300–2304. [PubMed: 35930748]
44. Andermann TM, Fouladi F, Tamburini FB, et al. A Fructo-Oligosaccharide Prebiotic Is Well Tolerated in Adults Undergoing Allogeneic Hematopoietic Stem Cell Transplantation: A Phase I Dose-Escalation Trial. Transplant Cell Ther. 2021;27: 932 e931–932 e911.
45. Stein-Thoeringer CK, Nichols KB, Lazrak A, et al. Lactose drives Enterococcus expansion to promote graft-versus-host disease. Science. 2019;366: 1143–1149. [PubMed: 31780560]
46. Gerbitz A, Schultz M, Wilke A, et al. Probiotic effects on experimental graft-versus-host disease: let them eat yogurt. Blood. 2004;103: 4365–4367. [PubMed: 14962899]

47. Gorshein E, Wei C, Ambrosy S, et al. Lactobacillus rhamnosus GG probiotic enteric regimen does not appreciably alter the gut microbiome or provide protection against GVHD after allogeneic hematopoietic stem cell transplantation. Clinical Transplantation. 2017;31: e12947.
48. Mathewson ND, Jenq R, Mathew AV, et al. Gut microbiome-derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease. Nat Immunol. 2016;17: 505–513. [PubMed: 26998764]
49. Eriguchi Y, Takashima S, Oka H, et al. Graft-versus-host disease disrupts intestinal microbial ecology by inhibiting Paneth cell production of α-defensins. Blood. 2012;120: 223–231. [PubMed: 22535662]
50. Severyn CJ, Siranosian BA, Kong ST, et al. Microbiota dynamics in a randomized trial of gut decontamination during allogeneic hematopoietic cell transplantation. JCI Insight. 2022;7.
51. Swimm A, Giver CR, DeFilipp Z, et al. Indoles derived from intestinal microbiota act via type I interferon signaling to limit graft-versus-host disease. Blood. 2018;132: 2506–2519. [PubMed: 30257880]
52. Wu K, Yuan Y, Yu H, et al. The gut microbial metabolite trimethylamine N-oxide aggravates GVHD by inducing M1 macrophage polarization in mice. Blood. 2020;136: 501–515. [PubMed: 32291445]
53. Michonneau D, Latis E, Curis E, et al. Metabolomics analysis of human acute graft-versus-host disease reveals changes in host and microbiota-derived metabolites. Nature Communications. 2019;10: 5695.
54. Gu Z, Xiong Q, Wang L, et al. The impact of intestinal microbiota in antithymocyte globulin-based myeloablative allogeneic hematopoietic cell transplantation. Cancer. 2022;128: 1402–1410. [PubMed: 35077579]
55. Haak BW, Littmann ER, Chaubard JL, et al. Impact of gut colonization with butyrate-producing microbiota on respiratory viral infection following allo-HCT. Blood. 2018;131: 2978–2986. [PubMed: 29674425]
56. Schluter J, Peled JU, Taylor BP, et al. The gut microbiota is associated with immune cell dynamics in humans. Nature. 2020;588: 303–307. [PubMed: 33239790]
57. Miltiadous O, Waters NR, Andrlova H, et al. Early intestinal microbial features are associated with CD4 T-cell recovery after allogeneic hematopoietic transplant. Blood. 2022;139: 2758–2769. [PubMed: 35061893]
58. Staffas A, Burgos da Silva M, Slingerland AE, et al. Nutritional Support from the Intestinal Microbiota Improves Hematopoietic Reconstitution after Bone Marrow Transplantation in Mice. Cell Host Microbe. 2018;23: 447–457 e444. [PubMed: 29576480]
59. Taur Y, Coyte K, Schluter J, et al. Reconstitution of the gut microbiota of antibiotic-treated patients by autologous fecal microbiota transplant. Sci Transl Med. 2018;10: eaap9489. [PubMed: 30257956]
60. Khan N, Lindner S, Gomes ALC, et al. Fecal microbiota diversity disruption and clinical outcomes after auto-HCT: a multicenter observational study. Blood. 2021;137: 1527–1537. [PubMed: 33512409]
61. Peled JU, Devlin SM, Staffas A, et al. Intestinal Microbiota and Relapse After Hematopoietic-Cell Transplantation. J Clin Oncol. 2017;35: 1650–1659. [PubMed: 28296584]
62. Blijlevens NM, Logan RM, Netea MG. The changing face of febrile neutropenia-from monotherapy to moulds to mucositis. Mucositis: from febrile neutropenia to febrile mucositis. J Antimicrob Chemother. 2009;63 Suppl 1: i36–40. [PubMed: 19372181]
63. Freifeld AG, Bow EJ, Sepkowitz KA, et al. Clinical practice guideline for the use of antimicrobial agents in neutropenic patients with cancer: 2010 update by the infectious diseases society of america. Clin Infect Dis. 2011;52: e56–93. [PubMed: 21258094]
64. Gibson GR, Roberfroid MB. Dietary modulation of the human colonic microbiota: introducing the concept of prebiotics. J Nutr. 1995;125: 1401–1412. [PubMed: 7782892]
65. David LA, Maurice CF, Carmody RN, et al. Diet rapidly and reproducibly alters the human gut microbiome. Nature. 2014;505: 559–563. [PubMed: 24336217]
66. Gurry T, Consortium* HSTM, Gibbons SM, et al. Predictability and persistence of prebiotic dietary supplementation in a healthy human cohort. Sci Rep. 2018;8: 12699. [PubMed: 30139999]

67. Ang QY, Alexander M, Newman JC, et al. Ketogenic Diets Alter the Gut Microbiome Resulting in Decreased Intestinal Th17 Cells. Cell. 2020;181: 1263–1275 e1216. [PubMed: 32437658]
68. Sorbara MT, Pamer EG. Microbiome-based therapeutics. Nat Rev Microbiol. 2022.
69. Keller JJ, Ooijevaar RE, Hvas CL, et al. A standardised model for stool banking for faecal microbiota transplantation: a consensus report from a multidisciplinary UEG working group. United European Gastroenterol J. 2021;9: 229–247.
70. DeFilipp Z, Bloom PP, Torres Soto M, et al. Drug-Resistant E. coli Bacteremia Transmitted by Fecal Microbiota Transplant. N Engl J Med. 2019;381: 2043–2050. [PubMed: 31665575]
71. van Prehn J, Reigadas E, Vogelzang EH, et al. European Society of Clinical Microbiology and Infectious Diseases: 2021 update on the treatment guidance document for Clostridioides difficile infection in adults. Clin Microbiol Infect. 2021;27 Suppl 2: S1–s21.
72. van Nood E, Vrieze A, Nieuwdorp M, et al. Duodenal infusion of donor feces for recurrent Clostridium difficile. N Engl J Med. 2013;368: 407–415. [PubMed: 23323867]
73. Baunwall SMD, Andreasen SE, Hansen MM, et al. Faecal microbiota transplantation for first or second Clostridioides difficile infection (EarlyFMT): a randomised, double-blind, placebo-controlled trial. Lancet Gastroenterol Hepatol. 2022;7: 1083–1091. [PubMed: 36152636]
74. Kelly CR, Ihunnah C, Fischer M, et al. Fecal Microbiota Transplant for Treatment ofClostridium difficileInfection in Immunocompromised Patients. American Journal of Gastroenterology. 2014;109.
75. Shono Y, Docampo MD, Peled JU, et al. Increased GVHD-related mortality with broad-spectrum antibiotic use after allogeneic hematopoietic stem cell transplantation in human patients and mice. Sci Transl Med. 2016;8: 339ra371.
76. Hayase E, Hayase T, Jamal MA, et al. Mucus-degrading Bacteroides link carbapenems to aggravated graft-versus-host disease. Cell. 2022;185: 3705–3719.e3714. [PubMed: 36179667]
77. Weber D, Jenq RR, Peled JU, et al. Microbiota Disruption Induced by Early Use of Broad-Spectrum Antibiotics Is an Independent Risk Factor of Outcome after Allogeneic Stem Cell Transplantation. Biol Blood Marrow Transplant. 2017;23: 845–852. [PubMed: 28232086]
78. Rashidi A, DiPersio JF, Westervelt P, et al. Peritransplant Serum Albumin Decline Predicts Subsequent Severe Acute Graft-versus-Host Disease after Mucotoxic Myeloablative Conditioning. Biol Blood Marrow Transplant. 2016;22: 1137–1141. [PubMed: 26988741]
79. Zama D, Gori D, Muratore E, et al. Enteral versus Parenteral Nutrition as Nutritional Support after Allogeneic Hematopoietic Stem Cell Transplantation: a Systematic Review and Meta-Analysis. Transplant Cell Ther. 2021;27: 180.e181–180.e188.
80. Andersen S, Staudacher H, Weber N, et al. Pilot study investigating the effect of enteral and parenteral nutrition on the gastrointestinal microbiome post-allogeneic transplantation. Br J Haematol. 2020;188: 570–581. [PubMed: 31612475]
81. D'Amico F, Biagi E, Rampelli S, et al. Enteral Nutrition in Pediatric Patients Undergoing Hematopoietic SCT Promotes the Recovery of Gut Microbiome Homeostasis. Nutrients. 2019;11.
82. Arends J, Bachmann P, Baracos V, et al. ESPEN guidelines on nutrition in cancer patients. Clin Nutr. 2017;36: 11–48. [PubMed: 27637832]
83. Iyama S, Sato T, Tatsumi H, et al. Efficacy of Enteral Supplementation Enriched with Glutamine, Fiber, and Oligosaccharide on Mucosal Injury following Hematopoietic Stem Cell Transplantation. Case Rep Oncol. 2014;7: 692–699. [PubMed: 25493082]
84. Yoshifuji K, Inamoto K, Kiridoshi Y, et al. Prebiotics protect against acute graft-versus-host disease and preserve the gut microbiota in stem cell transplantation. Blood Adv. 2020;4: 4607–4617. [PubMed: 32991720]
85. Upadhyaya B, McCormack L, Fardin-Kia AR, et al. Impact of dietary resistant starch type 4 on human gut microbiota and immunometabolic functions. Sci Rep. 2016;6: 28797. [PubMed: 27356770]
86. Venkataraman A, Sieber JR, Schmidt AW, Waldron C, Theis KR, Schmidt TM. Variable responses of human microbiomes to dietary supplementation with resistant starch. Microbiome. 2016;4: 33. [PubMed: 27357127]

87. van Lier YF, Davids M, Haverkate NJE, et al. Donor fecal microbiota transplantation ameliorates intestinal graft-versus-host disease in allogeneic hematopoietic cell transplant recipients. Sci Transl Med. 2020;12.

88. Malard F, Loschi M, Cluzeau T, et al. Pooled Allogenic Fecal Microbiotherapy MaaT013 for the Treatment of Steroid-Refractory Gastrointestinal Acute Graft-Versus-Host Disease: Results from the Phase IIa Heracles Study and Expanded Access Program. Blood. 2021;138: 262–262.

89. Qiao X, Biliński J, Wang L, et al. Safety and efficacy of fecal microbiota transplantation in the treatment of graft-versus-host disease. Bone Marrow Transplant. 2022.

90. DeFilipp Z, Peled JU, Li S, et al. Third-party fecal microbiota transplantation following allo-HCT reconstitutes microbiome diversity. Blood Adv. 2018;2: 745–753. [PubMed: 29592876]

91. Beelen DW, Elmaagacli A, Müller KD, Hirche H, Schaefer UW. Influence of intestinal bacterial decontamination using metronidazole and ciprofloxacin or ciprofloxacin alone on the development of acute graft-versus-host disease after marrow transplantation in patients with hematologic malignancies: final results and long-term follow-up of an open-label prospective randomized trial. Blood. 1999;93: 3267–3275. [PubMed: 10233878]

92. Beelen DW, Haralambie E, Brandt H, et al. Evidence that sustained growth suppression of intestinal anaerobic bacteria reduces the risk of acute graft-versus-host disease after sibling marrow transplantation. Blood. 1992;80: 2668–2676. [PubMed: 1421380]

93. Weber D, Jenq RR, Peled JU, et al. Microbiota Disruption Induced by Early Use of Broad-Spectrum Antibiotics Is an Independent Risk Factor of Outcome after Allogeneic Stem Cell Transplantation. Biol Blood Marrow Transplant. 2017;23: 845–852. [PubMed: 28232086]

94. Vossen JM, Guiot HF, Lankester AC, et al. Complete suppression of the gut microbiome prevents acute graft-versus-host disease following allogeneic bone marrow transplantation. PLoS One. 2014;9: e105706. [PubMed: 25180821]

95. Averbuch D, Orasch C, Cordonnier C, et al. European guidelines for empirical antibacterial therapy for febrile neutropenic patients in the era of growing resistance: summary of the 2011 4th European Conference on Infections in Leukemia. Haematologica. 2013;98: 1826–1835. [PubMed: 24323983]

96. Aguilar-Guisado M, Espigado I, Martin-Pena A, et al. Optimisation of empirical antimicrobial therapy in patients with haematological malignancies and febrile neutropenia (How Long study): an open-label, randomised, controlled phase 4 trial. Lancet Haematol. 2017;4: e573–e583. [PubMed: 29153975]

97. de Jonge NA, Sikkens JJ, Zweegman S, et al. Short versus extended treatment with a carbapenem in patients with high-risk fever of unknown origin during neutropenia: a non-inferiority, open-label, multicentre, randomised trial. Lancet Haematol. 2022;9: e563–e572. [PubMed: 35691326]

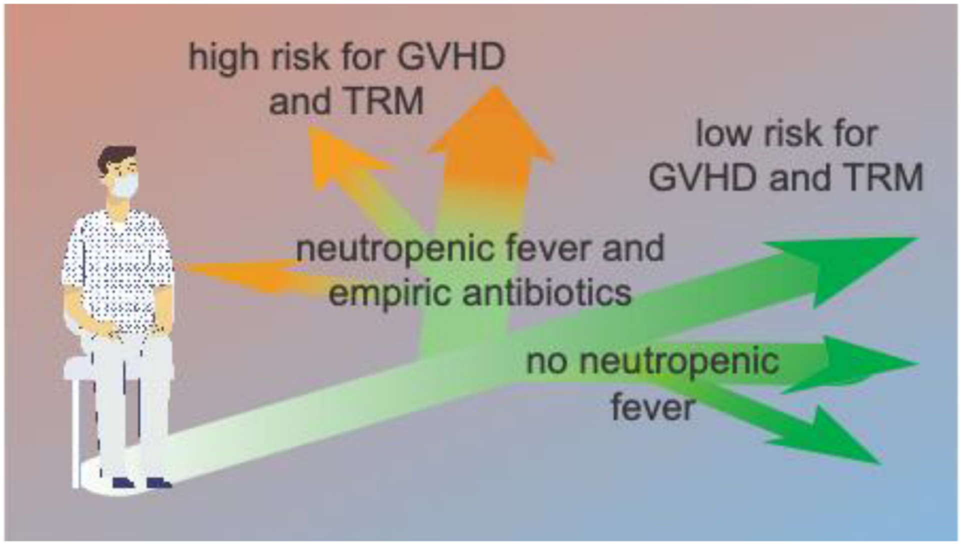

Figure 1.

Multiple studies indicate that use of antibiotics, including as empiric treatment of neutropenic fever, while necessary to prevent overwhelming infection, can also predispose to increased risk for adverse outcomes including GVHD and treatment-related mortality (TRM).

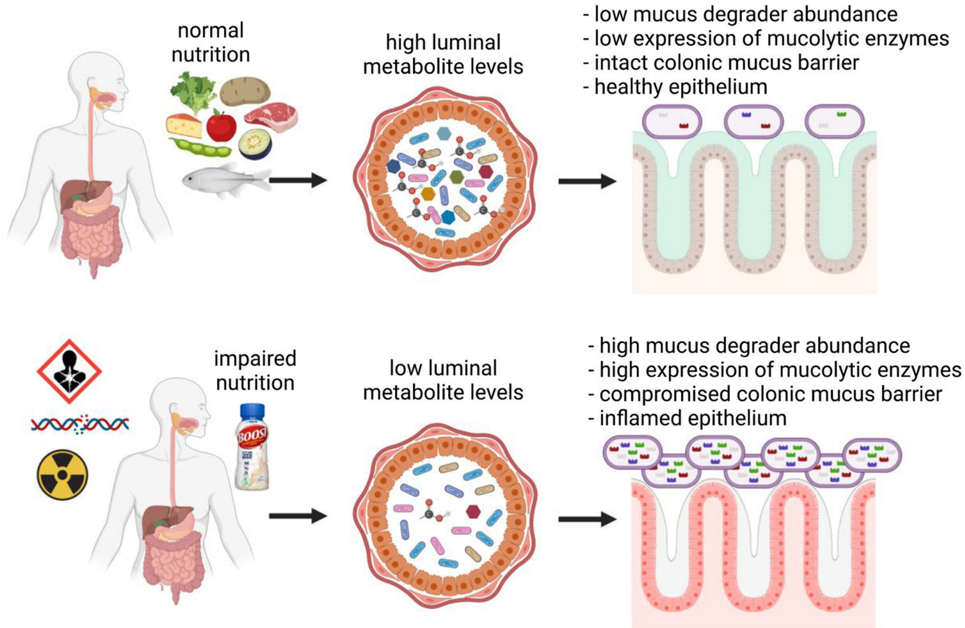

Figure 2.
Nutrition is an important modifier of the state of the intestinal microbiome. Impaired nutrition can lead to alterations in metabolite levels, which can in turn change the behavior of intestinal bacteria, including increasing the production of mucus-degrading enzymes by certain bacteria that are otherwise typically beneficial. Compromise of the colonic mucin barrier can in turn predispose to intestinal inflammation and translocation of microbial products into the systemic circulation, contributing to neutropenic fever. Created with BioRender.com.